Lost Teammates as Hazard Sensors: Stigmergic Marking of Robot Failures in a Foraging Swarm

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Abstract. Robots that explore unmapped environments are often lost to hazards their sensors cannot detect. Swarm robotics usually treats such losses as damage to be tolerated. We ask whether a decentralised foraging swarm with only local sensing can instead use the failures of its teammates as information about where the hazard is. In our controller, a robot that comes across the wreck of a teammate deposits a decaying repellent mark in a shared field, analogous to the "no entry" pheromone of Pharaoh ants, and robots steer down its gradient. The swarm uses site fidelity as its default search rule: after each delivery, a robot returns to where it last found food. We compare the marking controller with controllers that ignore wrecks, avoid only wrecks in view, add a risk trade-off, keep per-robot maps, use shared sighting records, or broadcast risk-limit alerts, in a simulated 20-robot central-place foraging task. Four hypotheses and their analysis were pre-registered, and runs are paired by random seed. Under site fidelity, marking delivered more items than ignoring the hazard (median paired difference +1 item of about 139; A12 = 0.556, Holm $p = 9.2\times10^{-4}$) and lost fewer robots than avoiding visible wrecks alone (A12 = 0.664, Holm $p = 1.0\times10^{-7}$). In the random-failure control the bound on marking's item cost was met (lower CI bound of A12 = 0.396, above the 0.36 margin), with a median paired difference of −5 items [−10, 1]. Risk weighting showed no detectable benefit. The nest-restart model (robustness) reproduced the same ordering. Risk-limit alerts that fired lost more robots than plain avoidance in both hazard conditions. The information medium (shared field, per-robot maps, sighting records) made no detectable difference. The effect held under two simultaneous hazards. In a pre-registered physics replication in Webots, with real robot bodies and collisions, marking again lost fewer robots than avoiding visible wrecks (A12 = 0.61, Holm $p = 8\times10^{-6}$), but its small gain in items delivered was not detectable. All code, configurations and raw results are released. Total: 38,600 runs.

Keywords: swarm robotics; stigmergy; foraging; fault tolerance; hazard localisation; virtual pheromones; pre-registration

1 Introduction

Robot swarms are proposed for search and rescue, mine clearance and planetary exploration: places where the environment is unknown and may be dangerous. Many dangers there are invisible to the cheap, short-range sensors that swarm robots carry: an unstable floor, a radiation source, soft ground, a region that interferes with electronics. Often the first sign of such a hazard is that a robot stops working.

Swarm robotics usually responds to failures with tolerance. A swarm is valued because it degrades gracefully: losing a few members does not stop the task. Fault-detection work asks whether robots can notice that a teammate has failed, in order to protect the swarm's health [2, 5]. But a failure that happens in a particular place is also a measurement of that place. If robots die where the hazard is, the positions of the dead robots form a map of the hazard that nobody had to sense directly.

Social insects use a related trick. Pharaoh ants mark unrewarding branches of their trail network with a repellent "no entry" pheromone [9], and repellent virtual pheromones have been studied in robot swarms [6]. Here we combine the two ideas. A robot that comes across a teammate's wreck deposits a decaying repellent mark around it, in the spirit of stigmergy (coordination through traces left in the environment), and every

robot steers away from strong marks. No robot plans, no robot holds a global map, and the hazard is never sensed directly.

We ask: *can a decentralised foraging swarm, using only local sensing, localise an unknown hazard from teammates failing near it, and how much foraging and how many robots does that save compared with swarms that merely tolerate failures?*

Our contributions are:

1. A minimal failure-marking controller for central-place foraging with site fidelity as the default search rule, plus six comparison controllers that each isolate one ingredient: memory, a risk trade-off, an oracle reference, two decentralised ways of holding the marks, and a literature-inspired risk-limit alert.
2. A pre-registered evaluation under site fidelity with statistics paired by seed, and a random-failure control that measures the cost of marking when wrecks carry no information about place. A robustness check repeats the key results under a nest-restart foraging model.
3. Exploratory sweeps over hazard placement, swarm size, lethality, hazard size, the information medium, detection noise, a moving hazard, risk-limit alert broadcasts, arena scale, and two simultaneous hazards.
4. A pre-registered physics-based replication in Webots, with differential-drive robot bodies and collisions, of the two main comparisons.
5. An open, reproducible pipeline: seeded independent random streams, provenance on every result row, and bit-exact regression tests across code versions.

We do not claim to be the first to treat robot loss as a hazard signal. Our claim is narrower: losses can be turned into decaying stigmergic marks that need no planning or global map, inside a decentralised foraging task, with an explicit trade-off between items delivered and robots lost.

2 Related work

Stigmergy and repellent pheromones. Repellent trail pheromones in Pharaoh ants [9] inspire our marking rule. In robot swarms, virtual pheromones are usually implemented as a shared field that agents read locally. Hunt, Jones and Hauert [6] tested the limits of pheromone stigmergy in high-density robot swarms and found that its benefit can degrade as density rises. This motivates our swarm-size sweep (Section 6.2).

Faults and failures in swarms. Work on fault detection asks how robots identify failed teammates [2, 5]. Hunt, Cullen and Hauert propose "value at risk" strategies, in which a robot broadcasts an alert to its neighbours when its accumulated exposure to a hazard exceeds a risk limit [7]. This is the closest analogue to our C7 controller, which we test and disclose failed to save robots beyond plain avoidance (Section 6.6). Neither the original nor C7 uses the place of a failure to map an environmental hazard.

Foraging and site fidelity. Collective foraging is a standard swarm-robotics task [14]. In ant-inspired central-place foraging algorithms, robots may return to the place where they last found food (site fidelity) instead of searching afresh from the nest [13]. Our main results use site fidelity as the default. Section 8 shows that a nest-restart model gives the same ordering.

Risk-aware multi-robot exploration. Aguirre et al. [1] study a swarm that chooses the safer of two areas from hazard event rates; it does no foraging and makes no inference from lost teammates. Srivastava, Suresh and Nieto-Granda [10] localise hazards with multiple failure-prone robots using behaviour-adaptive path planning. This is the closest work we know of, but it is centralised planning rather than stigmergic foraging.

To our knowledge, no previous study infers hazard location from lost teammates through purely local, decentralised marking during foraging, and quantifies both the items and the robots it saves.

3 Model

3.1 World

The world is a two-dimensional, continuous, walled arena of 10 m × 10 m. A circular nest of radius 0.75 m sits at the centre, and four circular item patches (radius 0.75 m, 10 items each) sit at (2.5, 2.5), (7.5, 2.5), (2.5, 7.5) and (7.5, 7.5) m (Fig. 1). A collected item reappears uniformly at random within its own patch, so the resource distribution is stationary. Time advances in steps of 0.1 s. A run lasts 10,000 steps, about 17 simulated minutes. For the arena-scale experiments (E11), a scale factor multiplies all distances (arena, nest, patch and hazard positions) while radii are unchanged; at scale 1 results are bit-identical to all other experiments.

A hazard is a disc that robots cannot see. Each step a robot spends inside it, the robot fails with probability p (the lethality). At baseline there is one hazard of radius 1 m with $p = 0.02$, placed next to the patch at (7.5, 7.5). A failed robot becomes a *wreck*: static, permanent, and detectable by any robot within the sensing range.

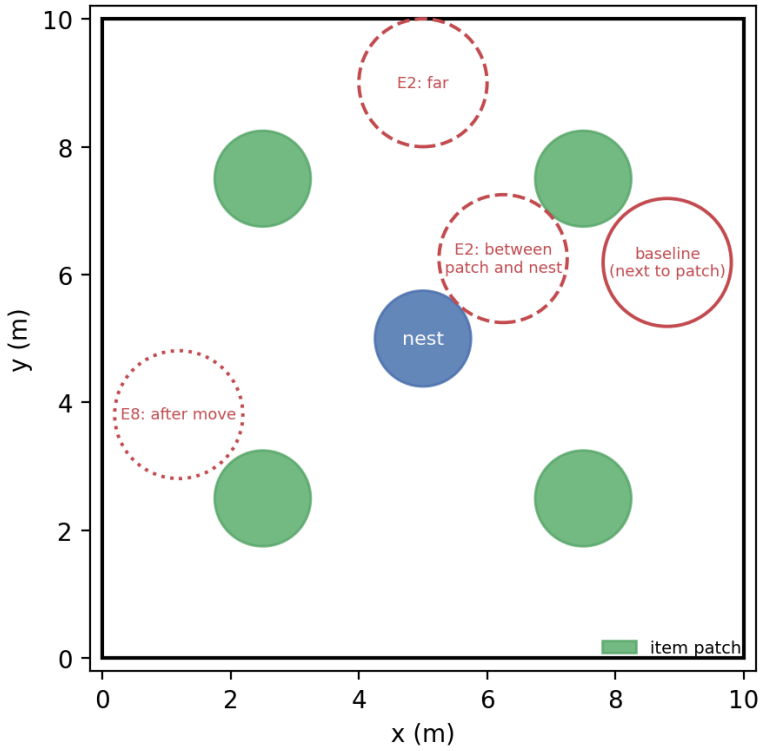


Fig. 1. The arena, with the nest (blue), four item patches (green) and the hazard positions used (red circles, radius 1 m). Solid: baseline position next to a patch. Dashed: E2/E2F's "between patch and nest" and "far" positions, and E12's second hazard. Dotted: the position the hazard moves to in E8/E8F.

3.2 Robots

Robots are point agents with an e-puck-like speed of 0.1 m/s and a sensing range of 0.5 m for items and wrecks. They always know the direction of the nest (as with a light beacon). There are no robot–robot collisions, a limitation discussed in Section 8. All controllers share one finite-state machine. In SEARCH a robot does a correlated random walk. When it senses an item it approaches and picks it up. In RETURN it

heads home, drops the item at the nest, and transitions back to the search phase. Under site fidelity (OUTBOUND) a robot that has previously picked up a food item heads straight to its last pickup position after each delivery, then switches to SEARCH within sensing range of it; no pickups occur in OUTBOUND. Under the nest-restart rule, a robot resumes SEARCH at the nest after each delivery. The controllers differ only in how they respond to wrecks and marks.

3.3 Controllers

ID	Name	Response to wrecks
C0	Ignore	None: wrecks do not affect behaviour.
C1	Avoid wrecks	Local repulsion from wrecks currently in view; no memory, no marking.
C2	Mark (main method)	C1 plus marking: the first time a robot sees a given wreck, it adds a Gaussian bump to a shared repellent grid (0.1 m cells) that decays every step. Robots steer down the field's local gradient in SEARCH, RETURN and OUTBOUND.
C3	Mark + risk trade-off	C2 with the repellent push scaled by $1/(1 + k \cdot n)$, where n is the number of items in view.
C4	Oracle	Cannot enter the hazard; slides along its edge. A reference, not a swarm method.
C5	Per-robot maps	Each robot marks its own map; every 1 s it takes the cell-wise maximum with the maps of live robots within communication range. Uses C2's parameters, untuned.
C6	Shared sighting records	Each robot keeps the sighting records it knows of, exchanging them by set union with robots within communication range every 1 s. Its map is the sum of decayed bumps over the records it knows, so with full propagation it equals C2's shared field. C2's tuned parameters; not tuned itself.
C7	Risk-limit alerts	From the same first-sighting evidence as C2, each robot accumulates a decaying risk score and, at the alert limit, broadcasts its position to live robots within communication range; receivers avoid that position for a fixed duration. No environmental marks; C1-style wreck avoidance. Adapted from Hunt et al. [7].

Table 1. Controllers. C1 versus C2 isolates the value of remembering where wrecks are. C2 versus C6 changes only where the information is held, and C6 versus C5 changes only how marks are combined (sum versus maximum). C7 is our adaptation of the risk-limit alert mechanism; it is not Hunt et al.'s method.

3.4 Detection noise and hazard movement

For the noise experiments, a robot misses a wreck in range with probability q_{miss}, either independently each step or in "blind spells" of a given mean length (correlated misses). Separately, each live robot has a "false sighting" of a non-existent wreck at its own position with probability q_{false} per step. The marking controllers deposit a mark at a false sighting. In the moving-hazard experiment, the hazard is replaced at step 5,000 by one at another position. Wrecks and marks stay where they are.

4 Methods

4.1 Metrics

The primary metric is *items delivered* to the nest per run. The secondary metrics are *robots lost*; *items per robot lost* (total items over total robots lost, per controller and condition); and the *localisation AUC* of the repellent field. The AUC is the area under the ROC curve that ranks grid cells by repellent value against the true hazard mask. It is 0.5 for an empty or uninformative field and 1.0 for a perfect one; for C5 and C6 it is computed on the mean of the live robots' maps.

4.2 Experiments

Exp.	Factor	Levels	Controllers	Runs
Site-fidelity model (main)				
E1F	Baseline	—	C0–C4	500
E2F	Hazard placement	next to patch; between patch and nest; far; none (random failures)	C0–C4	2,000
E3F	Swarm size	5, 10, 20, 40, 80 robots	C0–C4	2,500
E4F	Lethality p	0.005, 0.02, 0.1, 1.0 per step	C0–C4	2,000
E5F	Hazard radius	0.5, 1, 2 m	C0–C4	1,500
E6bF	Information medium	5–80 robots	C2, C5, C6	1,500
E7bF	Detection noise (strong)	none; miss 0.5; miss 0.8; correlated 0.5; correlated 0.8; $q_{\text{false}} = 10^{-4}$	C0–C3	2,400
E8F	Hazard moves at 500 s	static (decay 0.0002†); moved (decay 0.0002†); moved (decay 0.002)	C0, C1, C2, C4	1,200
Journal package (exploratory)				
E10	Risk-limit alerts	baseline; between patch and nest; random failures	C0, C1, C2, C7	1,200
E10b	Alerts that fire (alert_limit 1)	same three cells	C0, C1, C2, C7	1,200
E11	Arena scale; comm. range	n 20 comm 1 m; n 20 comm 0.5 m; n 20 comm 0.25 m; arena×2 n 80 comm 1 m; arena×2 n 80 comm 0.5 m	C2, C5, C6	1,500
E12	Two hazards	one hazard; two hazards; two hazards + $p = 0.1$	C0–C4	1,500
Nest-restart model (robustness; Section 8)				
E1–E5	Baseline, placement, size, lethality, radius	(as E1F–E5F)	C0–C4	8,500
E6, E6b	Information medium	5–80 robots	C2, C5 (E6); C2, C5, C6 (E6b)	2,500
E7, E7b	Detection noise	mild (E7); strong & correlated (E7b)	C0–C3	4,000
E8	Hazard moves	static; moved (decay 0.0002); moved (decay 0.002)	C0, C1, C2, C4	1,200
E9	Search restart rule	nest vs fidelity × baseline / random-fail / false-sight	C0–C4	3,000
W1	Physics replication (Webots)	site-fidelity baseline	C0, C1, C2, C4	400

Table 2. Experiments. Each condition (cell) has 100 seeds; every controller in a cell runs on the same seeds. All site-fidelity and journal experiments use `configs/tuned_params_fidelity.yaml` and have paired cells. E1–E9 (nest model) use `configs/tuned_params.yaml`. † E8F cells 0 and 1 used decay = 0.0002 (the nest-tuned value), not the fidelity-tuned 0.002; see Appendix A. W1 runs in Webots (Section 7). Grand total: 38,600 runs.

4.3 Random-failure control

To measure the cost of marking when wrecks carry no information about place, E2F's fourth cell has no hazard. Instead, every live robot fails independently with probability $q = 2.66\times10^{-5}$ per step, anywhere in the arena. This value matches C0's mean baseline losses in a pilot (4.67 of 20 robots per run). Random failures use their own random stream, so all other randomness is the same as in runs without them.

4.4 Parameter tuning

C2's repulsion weight, deposit width and decay rate, and C3's risk constant k, were tuned separately for each foraging model by grid search at the baseline configuration, on seeds disjoint from all main experiments, using a pre-registered rule: highest median items delivered, with ties broken by lowest mean robots lost.

Under site fidelity (tune_c2f, tune_c3f): the chosen values are repulsion weight 4.0, deposit width (σ) 0.3 m, decay 0.002 per step and k = 0.5. The top tuning cell gave 146.0 median items against 143.5 for the second-best, so the landscape was less flat than under nest restart (see below). All F experiments and journal experiments load these values from `configs/tuned_params_fidelity.yaml`.

Under nest restart (tune_c2, tune_c3): the chosen values are repulsion weight 0.5, deposit width 0.3 m, decay 0.0002 per step and k = 0.5. The spread of cell means (SD 1.09 items) matched what seed noise alone predicts (SE 1.13), so C2's performance under nest restart was insensitive to these parameters over the grid tested.

C5 and C6 use C2's tuned parameters; C7 was tuned on a separate grid (`tune_c7f`; see Section 6.6).

4.5 Statistics

All comparisons are paired by seed. The confirmatory tests are one-sided Wilcoxon signed-rank tests on per-seed differences [11], with zero differences handled by Pratt's method [8], and Holm correction [4] across H1F, H2F and H4F at $\alpha = 0.05$. H3F is a non-inferiority bound, decided only by the bootstrap confidence interval. Effect sizes are Vargha–Delaney A12 [12]: the probability that a run of the first controller scores higher than a run of the second, with ties counting half (0.5 means no difference). Confidence intervals are 95% percentile bootstrap intervals [3] with 10,000 resamples, resampling seeds so that pairing is kept. Exploratory analyses use a Friedman test per cell, two-sided Wilcoxon tests against C2, and Holm correction within each experiment. We report "no detectable difference" rather than equivalence.

4.6 Pre-registration and deviations

The hypotheses, metrics, statistics and experiment levels were committed to the project repository before any main experiment ran. The site-fidelity replication (H1F–H4F, E1F–E8F) was pre-registered as a section of the same document before any F experiment ran. Later experiments (E10–E12, E10b) were pre-registered before their data existed. Deviations are listed in Appendix A. The most important are: a pilot go/no-go check narrowly missed its threshold and we proceeded by a recorded decision; the confirmatory test changed from Mann–Whitney U to paired Wilcoxon before the main experiments; the week-7 replication was not blind (E9 had already shown its result in the fidelity cells); E8F cells 0 and 1 used an incorrect decay value; and E10b was designed after seeing E10 data (disclosed in the pre-registration section for E10b).

4.7 Reproducibility

Each run draws from independent random streams (layout, motion, item respawn, hazard, random failures, noise), seeded by experiment, cell and seed. Every result row records the code commit, package versions and import path. Experiments run after code changes ran from a frozen copy of the repository. They were accepted only if every regression ("golden") test case reproduced bit-identically, which pins the behaviour of C0–C5 across versions. The golden cases reproduced bit for bit on a second Windows machine (the GitHub Actions *windows-latest* runner). On Linux (*ubuntu-latest*), 19 of the 20 golden cases matched in every recorded result value and differed only in the final-state fingerprint (floating-point rounding); the twentieth differed by 2 robot-steps in 25,297 in one observation metric. We therefore claim exact re-runs on Windows and agreement to within floating-point rounding elsewhere.

5 Results: confirmatory replication under site fidelity

The hypotheses H1F–H4F are replications of H1–H4 under site fidelity. The analysis and test family are the same; only the foraging model and tuned parameters differ. We note the non-blindness of this replication: E9 (week 6) already showed, in its fidelity cells, that C2 delivered more items than C0 (Holm $p = 0.026$ within E9) and lost fewer robots than C1 (Holm $p < 10^{-8}$).

	Claim	Medians	Paired diff. [95% CI]	A12 [95% CI]	Holm p	Verdict
H1F	C2 delivers more items than C0 (E1F)	139 vs 135	+1 [0, 4]	0.556 [0.524, 0.592]	9.2×10^{-4}	Supported
H2F	C2 loses fewer robots than C1 (E1F)	2 vs 3	+1 [0, 1]	0.664 [0.609, 0.719]	1.0×10^{-7}	Supported
H3F	C2's item cost vs C0 is small with random failures (E2F)	127 vs 124.5	−5 [−10, 1]	0.455 [0.396, 0.513]	CI rule	Supported
H4F	C3 delivers more items than C2 (E1F)	139 vs 139	0 [0, 0]	0.499 [0.479, 0.518]	0.453	Not supported

Table 3. Confirmatory results under site fidelity (100 seeds each). The paired difference is the first-named controller minus the second (C1 − C2 for H2F on robots lost). H3F is supported if the lower bootstrap bound of A12(C2, C0) exceeds 0.36; the bound is 0.396. The paired difference for H3F is negative (C2 delivered 5 fewer items on the median seed), but the A12 bound remains met.

H1F: marking helps foraging, modestly. C2 delivered a median of 1 more item per seed than C0, out of about 139. The effect is smaller than under nest restart (+3 items of about 45) because site fidelity tripled throughput for all controllers, compressing the gap between them.

H2F: marking saves robots beyond what avoidance does. C2 lost fewer robots than C1 on most seeds. C1 avoids a wreck only while it is within 0.5 m; C2's mark stays in place and covers the area around the wreck. The robot savings under site fidelity (median 1 robot) matched those under nest restart.

H3F: bound met, but no advantage. With no hazard and random failures scattered anywhere, marking neither helps nor noticeably hurts foraging. The lower bound of A12(C2, C0) = 0.396 exceeds 0.36, so H3F is supported. Unlike under nest restart, however, there is no positive lead for C2 (median paired difference −5 items). With site fidelity, the foraging model no longer generates the near-nest inflation that gave C2 a spurious advantage in the random-failure control (see Section 7.1).

H4F: risk weighting showed no detectable benefit. C3 delivered exactly the same number of items as C2 on almost all seeds. This is not evidence that a risk trade-off is useless at other settings.

In the descriptive E1F results, the median items delivered by C0–C4 were 135, 138, 139, 139 and 148, and the median robots lost were 4, 3, 2, 2 and 0 (Fig. 2). Items per robot lost rose from 36.1 (C0) and 50.5 (C1) to 64.5 (C2). On items delivered, C2 closes about one-third of the gap between C0 and the oracle C4 (13 items total gap; C2 gains 4 over C0).

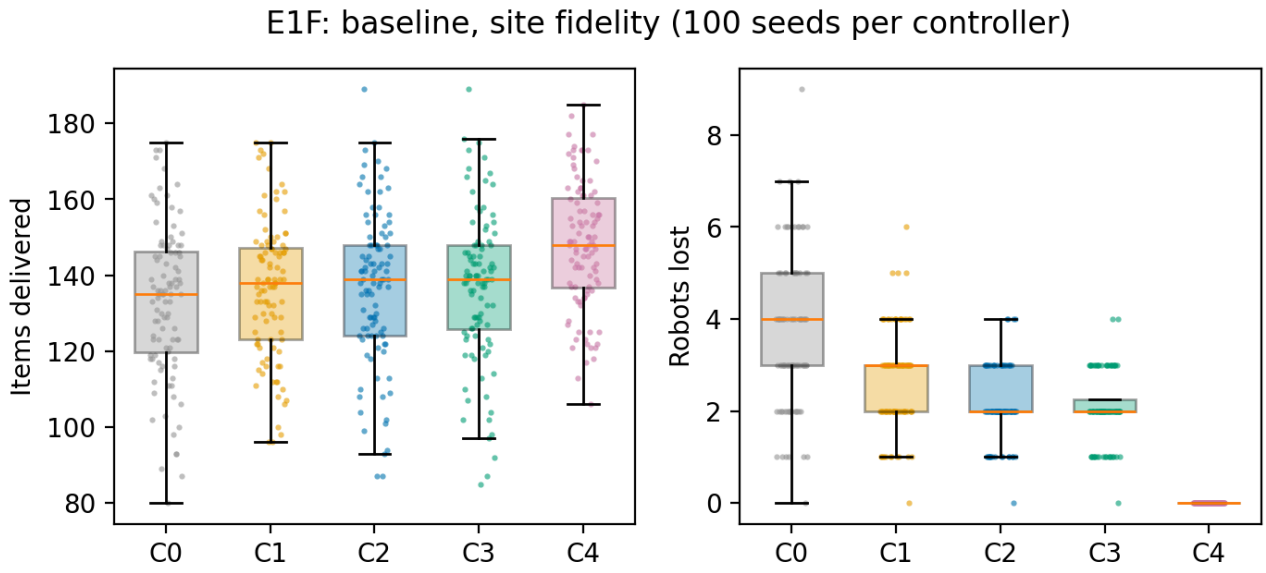


Fig. 2. Baseline under site fidelity (E1F, 100 seeds). Items delivered (left) and robots lost (right) for C0–C4; each dot is one seed.

5.1 Original hypotheses under the nest-restart model

For comparison, Table 4 shows the pre-registered confirmatory results under the nest-restart foraging model (E1). The direction and significance of H1–H3 are the same as in the site-fidelity replication; H4 is also not supported. Under nest restart the item gain for H1 is larger (+3 vs +1) and the A12 values for both H1 and H2 are larger, but this reflects the greater variability in a less efficient foraging model rather than a fundamentally different effect.

	Claim	Medians	Paired diff. [95% CI]	A12 [95% CI]	Holm *p*	Verdict
H1	C2 delivers more items than C0 (E1)	45 vs 42	+3 [1.5, 4]	0.661 [0.620, 0.706]	5.0×10^{-11}	Supported
H2	C2 loses fewer robots than C1 (E1)	2 vs 3	+1 [1, 1]	0.789 [0.742, 0.835]	1.7×10^{-14}	Supported
H3	C2's item cost vs C0 with random failures (E2)	45 vs 43	+1 [0, 2.5]	0.582 [0.519, 0.646]	CI rule	Supported
H4	C3 delivers more items than C2 (E1)	45 vs 45	0 [0, 0]	0.503 [0.499, 0.510]	0.156	Not supported

Table 4. Confirmatory results under nest restart (100 seeds each). H3: lower bootstrap bound of A12(C2, C0) = 0.519 > 0.36.

6 Exploratory results under site fidelity

Everything in this section is exploratory: tests are two-sided Wilcoxon tests against C2 with Holm correction within each experiment, and explanations are labelled as candidates unless the numbers show them directly.

6.1 Hazard placement (E2F)

Placement mattered most when the hazard sat between a patch and the nest, on a path the swarm uses constantly (Fig. 3). In that cell, C0 delivered a median of 76 items with 12 of 20 robots lost, while C2 reached 128.5 items and lost 5. The gap between C0 and C2 was larger here than at baseline or in the far-

hazard cell. C2 delivered more than C0 in all three hazard cells (Holm $p < 10^{-3}$). The model-comparison table (Section 8.2) shows that the pattern mirrors E2 closely.

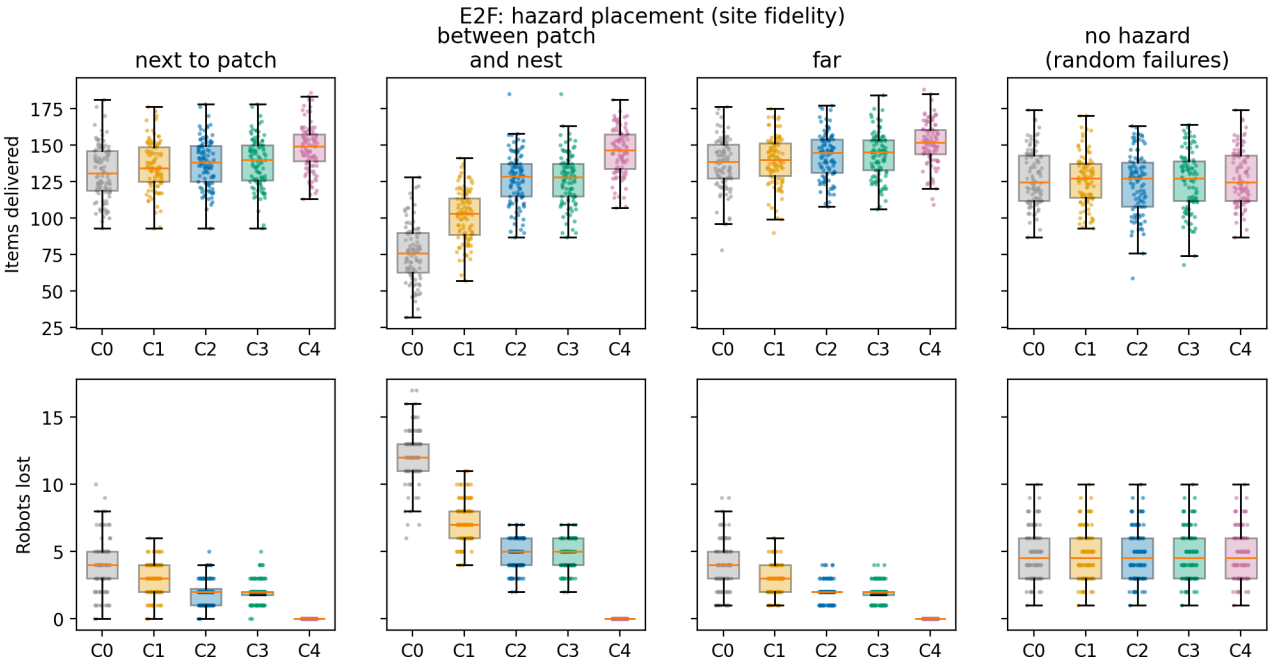


Fig. 3. Hazard placement under site fidelity (E2F). Items delivered (top) and robots lost (bottom) for each placement. In the random-failure cell, robots lost is nearly identical across controllers by design.

6.2 Swarm size (E3F)

The robots saved by marking grew with swarm size (Fig. 4). The median robots lost at 5, 10, 20, 40 and 80 robots were 1, 2, 4, 8 and 15 for C0; 1, 2, 3, 4 and 6 for C1; and 1, 1, 2, 3 and 3 for C2. At 80 robots, C2 delivered 512.5 items against C0's 480, and items per robot lost were 161.8 against 31.1. At 5 robots there was no detectable difference. Unlike the density-related degradation of pheromone stigmergy reported by Hunt et al. [6], we saw no degradation up to 80 robots in this arena.

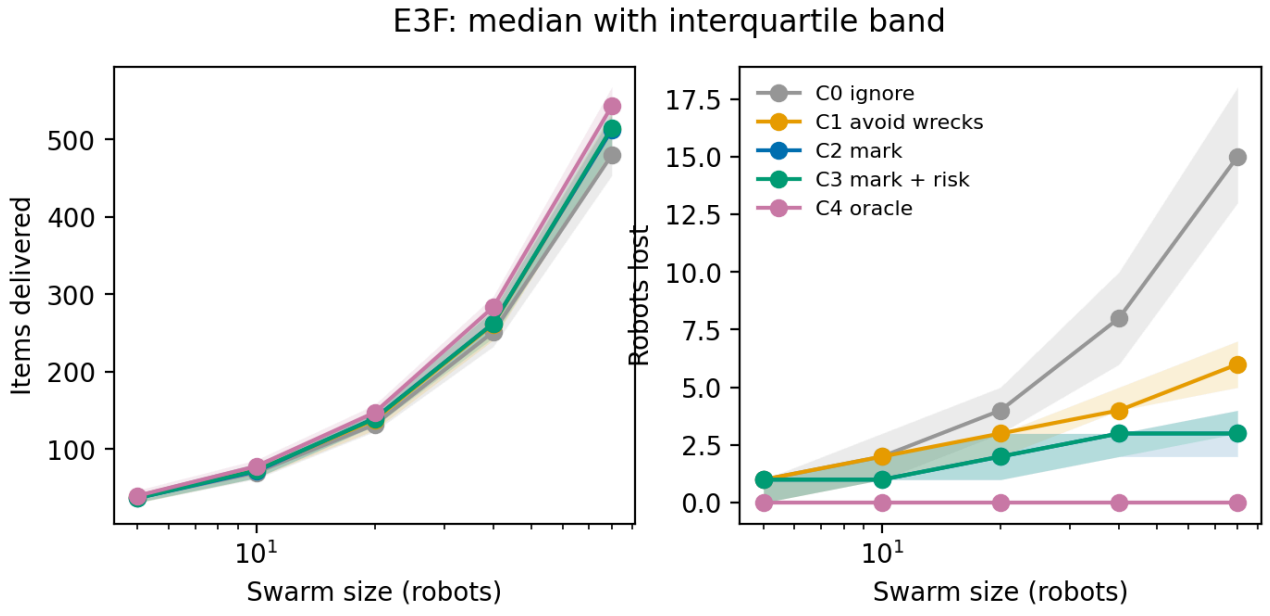


Fig. 4. Swarm size under site fidelity (E3F): medians with interquartile bands against the number of robots (log axis).

6.3 Lethality (E4F)

C2 was nearly flat across lethality from 0.005 to 1.0 per step, delivering 135–142.5 items and losing 2 robots at every level. C0's losses rose from a median of 3 (at $p = 0.005$) to 4 (at $p = 0.02$ and 0.1) and stayed there, reflecting the same saturation pattern as under nest restart: above $p \approx 0.1$, nearly every robot that enters the hazard is lost, so losses are limited by how many robots enter rather than by lethality.

6.4 Hazard size (E5F)

Against the hazard's in-arena area (0.75, 3.18 and 8.23 m²), C0's median losses rose from 2 to 4 to 6 and deliveries fell from 134.5 to 128 to 118. C2 lost 1, 2 and 3 robots and delivered 136, 133 and 129 items. The oracle delivered 145.5–150.5 across all radii. C2's localisation AUC fell with area (comparable to E5 under nest restart), consistent with marks clustering at the hazard's edge.

6.5 Information medium (E6bF)

Per-robot maps (C5) and shared sighting records (C6) showed no detectable difference from the shared field (C2) in any combination of swarm size and experiment under site fidelity (all Holm $p \geq 0.52$; E11 further confirms this at reduced communication ranges; see Section 6.7). The median fraction of sighting records known at run end was high across all sizes (E6bF), consistent with rapid propagation in this arena.

6.6 Risk-limit alerts (E10, E10b)

C7 implements the alert mechanism inspired by Hunt et al. [7]: a robot accumulates a decaying risk score from wreck sightings and broadcasts its position when the score reaches a limit. We pre-registered a tuning grid over alert_limit {1, 2, 3} and alert_duration {300, 1000, 3000}. **The grid was poorly designed.** With alert_decay = 0.001 a robot's score rarely reached 2 or 3: mean alerts per run were 0.00–0.10 at those limits. The pre-registered selection rule chose alert_limit = 3 on noise, so C7 in E10 was effectively C1.

E10 is reported as run (Fig. 5, left panel). In the between-patch-and-nest cell, C7 delivered 106 items and lost 7 robots, identical to C1, while C2 delivered 129 items and lost 4. In the baseline cell C7 lost 3 robots and delivered 134.5, also matching C1. C7 vs C1 on robots lost: A12 = 0.5, Holm $p = 1.0$ in every cell.

E10b repeated E10 with alert_limit = 1, the only grid value where alerts fired, and alert_duration = 3000 (the best duration at that limit; median 137.5 items in tuning). This choice was made after seeing E10 and tune_c7f and is disclosed as such (Appendix A). **Alerts that fired lost more robots than plain avoidance.** In the between-patch-and-nest cell, C7 lost a median of 9 robots against C1's 7 (A12(C7, C1) on robots lost = 0.754, Holm $p = 1.5\times10^{-8}$). In the baseline cell, C7 lost a median of 3 against C1's 3, but A12 = 0.618 (Holm $p = 3.2\times10^{-5}$); the excess loss was smaller but detectable (Fig. 5, right panel). In the random-failure cell there was no detectable difference. C2 delivered more items and lost fewer robots than C7 in every hazard cell. A candidate explanation is that alert broadcasts cause receivers to avoid an area for a fixed duration regardless of whether new robots are dying there; this may divert robots from paths near the hazard that C2's field would permit once no fresh marks appear.

We emphasise that C7 is our adaptation, not Hunt et al.'s method: we derived it from the same first-sighting evidence as C2's marks, whereas Hunt et al. use a separate exposure accumulator with a different trigger. The comparison shows that our adaptation of the alert concept, as implemented here, did not reduce robot losses.

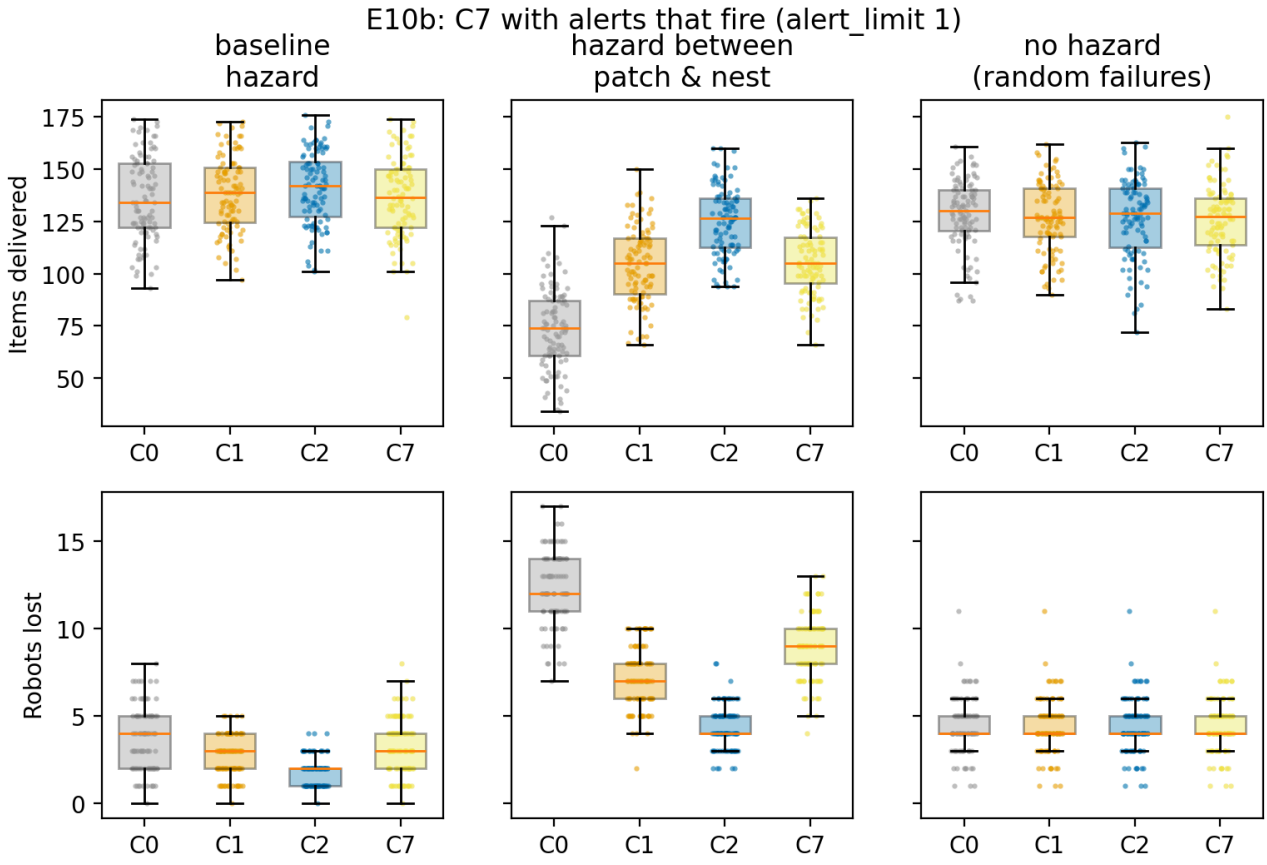


Fig. 5. Risk-limit alerts. Left: E10 (tuned C7, effectively C1). Right: E10b (alert_limit = 1, alerts fire). Items delivered (top) and robots lost (bottom). In E10b's between-patch-and-nest cell, C7 lost more robots than C1.

6.7 Arena scale and communication range (E11)

E11 tested whether the null result of E6bF (no detectable difference between information media) holds at reduced communication ranges and in a larger arena. At three communication ranges (1, 0.5, 0.25 m) with 20 robots in the standard arena, and at two ranges (1, 0.5 m) with 80 robots in a 2×-scaled arena, C5 and C6 showed no detectable difference from C2 (all Holm p = 1.0; Fig. 6). At the smallest range (0.25 m, 20 robots), C6's median known fraction of sighting records fell to around 0.5 by run end, yet its deliveries and robot savings were statistically indistinguishable from C2. At arena scale 2 with 80 robots, all three controllers delivered about 52 items (the arena is 4× larger by area), with 2 robots lost.

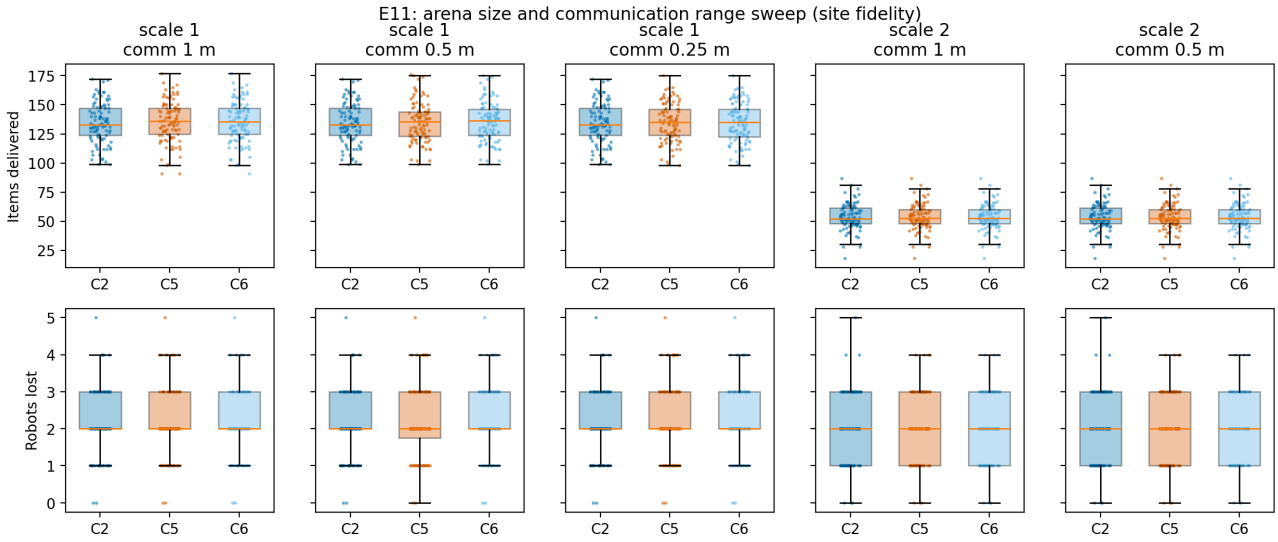


Fig. 6. Arena scale and communication range (E11): medians with bootstrap CIs for C2, C5 and C6 across the five cells.

6.8 Detection noise (E7bF)

Under strong and correlated detection misses (E7bF), C2 still lost fewer robots than C0 and C1 in every cell. The false-sighting gain seen under nest restart (C2 delivering 6–7 more items with frequent false sightings) was much smaller under site fidelity: a median of about 2 more items in E7bF's false-sighting cell (C2 delivered 141 items, against 139 with no noise). This confirms E9's interaction test: with site fidelity, false sightings no longer push robots away from the time-wasting near-nest region as effectively (Fig. 7).

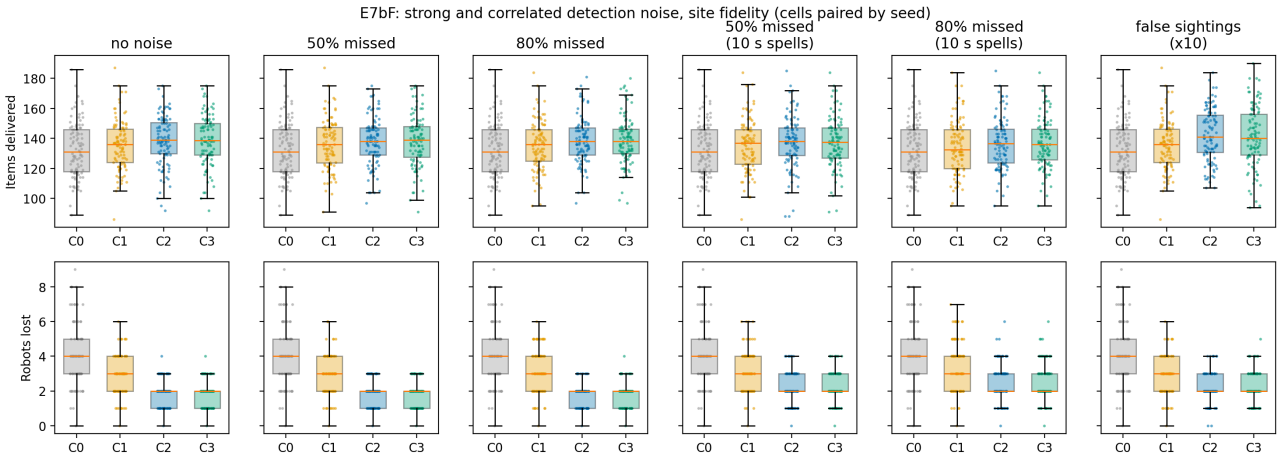


Fig. 7. Strong and correlated detection noise under site fidelity (E7bF; all cells share layouts). C2 consistently saves robots in every noise condition.

6.9 Moving hazard (E8F)

In E8F the hazard moves at 500 s (step 5,000) to the opposite side of the nest (Fig. 8). An important deviation: E8F cells 0 (static) and 1 (moved) used decay = 0.0002, the nest-tuned value, rather than the fidelity-tuned 0.002; only cell 2 used the correct 0.002 (Appendix A).

In the second half of the run, C2 lost a median of 0 robots when the hazard stayed (cell 0) and 2 when it moved (cell 1); it delivered 81.5 and 77 items (moved vs static: A12 = 0.88 on robots lost and 0.40 on items, both Holm $p < 10^{-4}$). After a move it still lost fewer robots than C0 (2 vs 3; Holm $p < 10^{-4}$) and C1 (A12 = 0.61, Holm $p = 0.0006$). The faster decay of cell 2 (0.002, the site-fidelity tuned value; cells 0 and 1 used

0.0002, copied from E8) showed no detectable difference from the slow decay (delivered 79.5 vs 77, Holm p = 0.92). The oracle lost no robots after the move.

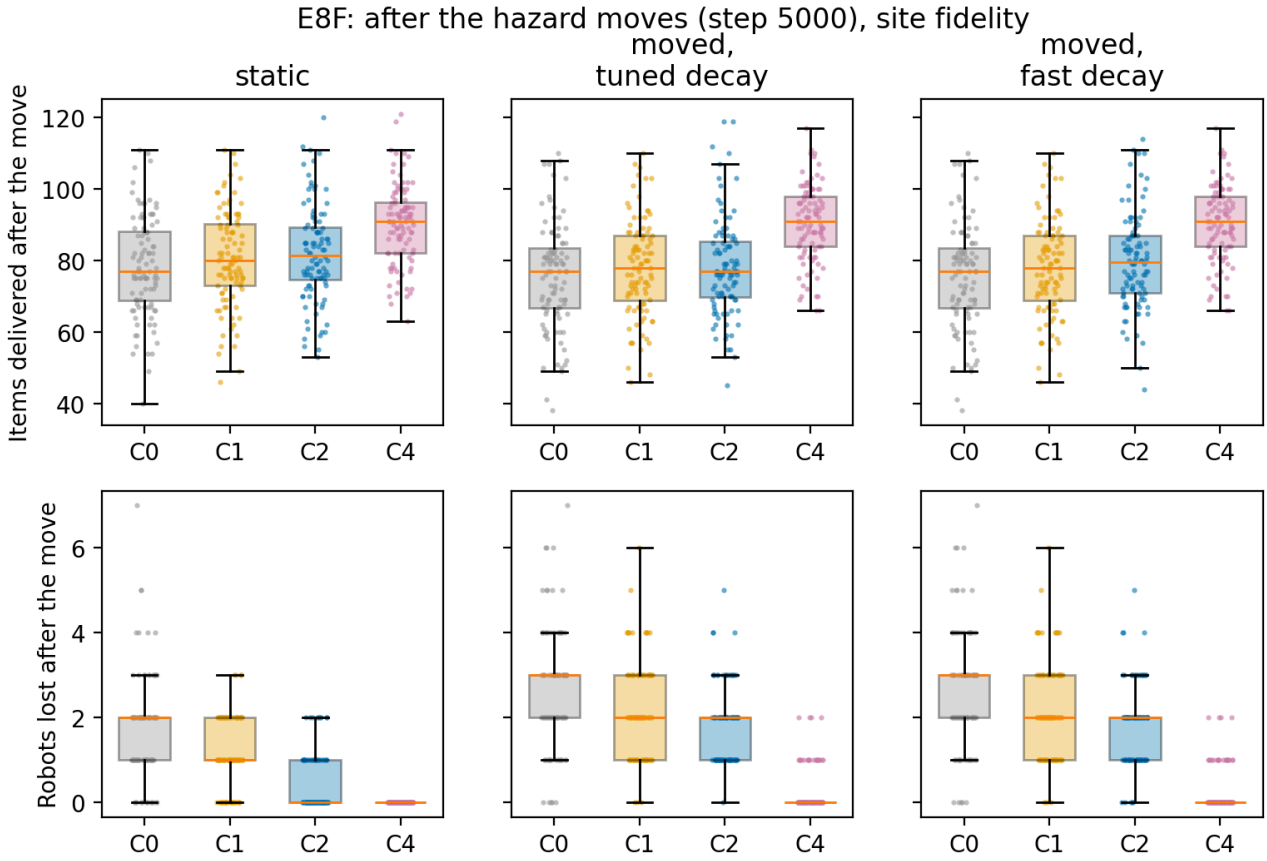


Fig. 8. Moving hazard under site fidelity (E8F). E8F cells 0 and 1 used the nest-tuned decay (0.0002); cell 2 used the fidelity-tuned decay (0.002).

6.10 Two simultaneous hazards (E12)

E12 placed a second hazard at (1.19, 3.81), next to another patch, alongside the standard baseline hazard. With two hazards, C2 delivered 128 items median and lost 4 robots, against C0's 120 items and 7 lost, and C1's 122.5 items and 5 lost (Fig. 9). C2 lost fewer robots than both and delivered more than C0 (Holm $p <$ 0.001); its lead over C1 in items was not detectable (Holm p = 0.082). With two hazards at high lethality (p = 0.1), C0 lost 8 robots and delivered 118 items, while C2 lost 4 and delivered 128. In all two-hazard cells, C2's localisation AUC (against both hazards) was above 0.75, and the oracle delivered around 150 items. These results show that the marking mechanism generalises to multiple simultaneous hazards without modification.

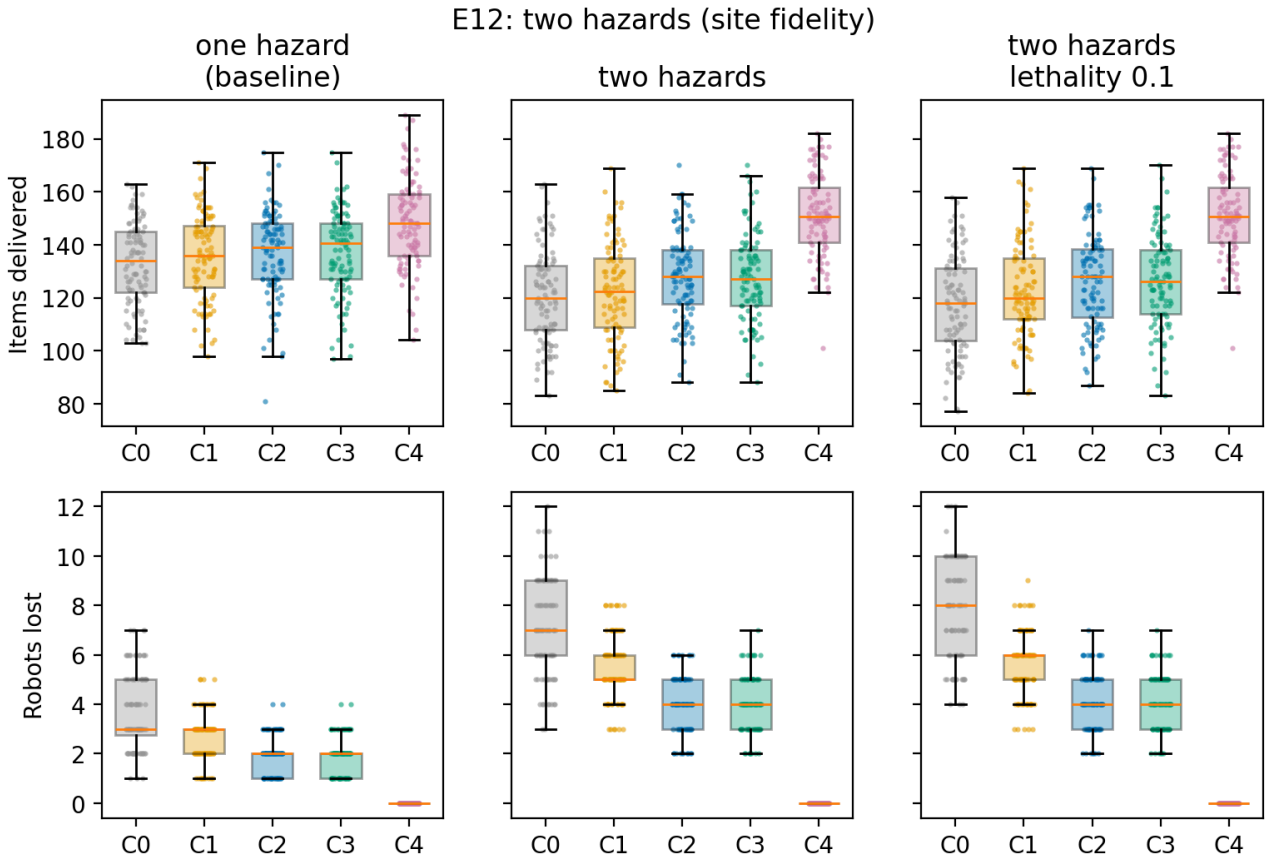


Fig. 9. Two simultaneous hazards (E12): items delivered (left) and robots lost (right) across the three cells.

7 Physics replication in Webots (W1)

All results above come from our own simulator, in which robots are points that never collide. To check that the main effects do not depend on that simplification, we replicated the site-fidelity baseline in Webots R2025a, an open-source robot simulator with rigid-body physics. Twenty differential-drive robots (7.4 cm body, 0.1 m/s) collide with the walls, with each other and with wrecks, which are stopped robots and so physical obstacles. A Webots supervisor provides what our robots sense virtually in every model: the hazard and its lethality, items and patches, the nest direction, wreck sensing within 0.5 m, the shared repellent field (with the site-fidelity tuned parameters) and site fidelity. The oracle C4 cannot be blocked physically, so it steers along the hazard edge and heads straight out if pushed in. A calibration without a hazard, run before the experiment, showed that real bodies slow foraging: C0 delivered 16–23 items in 3,000 steps against 19–37 in the 2D model. The experiment, W1, ran C0, C1, C2 and C4 for 100 seeds each, with the same layout and start positions for every controller on a given seed, and its two hypotheses were pre-registered before any W1 data existed.

	Claim	Medians	A12 [95% CI]	Holm p	Verdict
H1W	C2 delivers more items than C0	55.5 vs 55	0.486 [0.450, 0.519]	0.73	Not supported
H2W	C2 loses fewer robots than C1	1 vs 2	0.608 [0.566, 0.654]	8×10^{-6}	Supported

Table 5. Webots replication (100 seeds per controller, paired by seed). For H2W, A12 is A12(C1, C2) on robots lost.

The robot savings survive physics; the item gain does not. Marking lost fewer robots than avoiding visible wrecks: C2 lost fewer on 32 seeds, the same number on 63 and more on 5. On items delivered, C2 showed no detectable difference from C0 (C2 delivered more on 22 seeds, the same on 52 and fewer on 26). Physics

made foraging much slower than in our simulator (C0 delivered a median of 55 items against 135 in E1F, a descriptive comparison across simulators) and fewer robots died (C0 lost a median of 2 against 4). With fewer losses there is little to gain on items: even the oracle, which never entered the hazard, delivered a median of 57.5 items against C2's 55.5. A candidate explanation for the null H1W, not tested, is therefore that the item benefit is too small to detect when robots move slowly and die rarely. C2's map still pointed at the hazard (median final AUC 0.72; 60 of 100 runs had at least one mark).

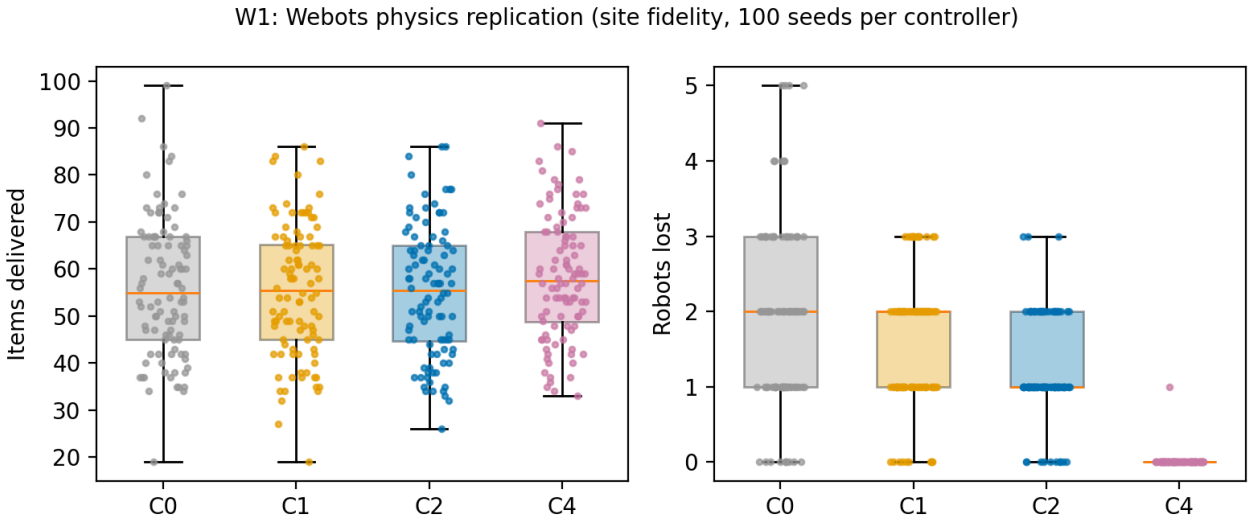


Fig. 10. Webots physics replication (W1): items delivered (left) and robots lost (right), each dot one seed.

8 Robustness: the nest-restart model

The experiments E1–E9 used search restarting at the nest after each delivery, the model in which the marking mechanism was originally developed. We treat them as a robustness check: do the site-fidelity results hold under a different foraging model?

8.1 Key results

Table 4 shows the confirmatory results (H1–H4). The direction of every finding matches the site-fidelity replication. In the exploratory sweeps, the same ordering held: C2 lost fewer robots than C0 and C1 across placements, swarm sizes, lethality levels, hazard sizes, noise conditions and a moving hazard. The absolute item counts were much lower (roughly one-third of the site-fidelity values) because nest-restart search is inefficient: robots spend about 38% of their steps within 1.5 m of the nest, compared with much less under site fidelity.

Under nest restart, marking showed an apparent advantage in the random-failure control (C2 +1 item over C0, lower A12 bound 0.519). A post-hoc diagnostic linked this to a side effect of the foraging model: because random wrecks fall disproportionately near the nest (where robots spend most time), C2's marks push robots outward from the nest, shortening their random walk back to food. E9 confirmed this: with site fidelity, the advantage disappeared (Holm $p = 1$ in E9's random-failure fidelity cell). We therefore read H3 strictly as a bound on cost, not as a benefit of marking.

Under nest restart, false sightings (E7b cell 5, $q_{\text{false}} = 10^{-4}$) gave C2 a large spurious gain (paired median +7 items over no noise). E9 showed this gain fell to about 2 items under site fidelity, confirming the nest-effect explanation.

8.2 Model comparison

Table 6 shows selected medians from the site-fidelity and nest-restart models side by side. Under site fidelity, C2 consistently delivers about three times as many items (site fidelity accelerates all controllers equally), while robot savings are comparable in absolute terms. The model comparison is descriptive only: the runs used different random streams and cannot be paired.

Exp.	Controller	Items (fidelity)	Lost (fidelity)	Items (nest)	Lost (nest)
E1F / E1	C0	135	4	42	5
	C1	138	3	43	3
	C2	139	2	45	2
	C4	148	0	48	0
E2F(1) / E2(1)	C0	76	12	17	17
(between patch & nest)	C2	128.5	5	42	4.5
E3F(4) / E3(4)	C0	480	15	150	19
(80 robots)	C2	512.5	3	164.5	3

Table 6. Selected medians from the model comparison (descriptive only; models are unpaired). Cell notation: (cell index) in parentheses.

9 Discussion

Failures carry information, and using it mainly saves robots. The largest and most consistent effect of marking is on robots lost: H2F at baseline, and in every placement, size, noise and movement condition under site fidelity, and matching results under nest restart. The gain in items delivered is modest at baseline (+1 of about 139 under site fidelity), but it grows when the hazard lies on a path the swarm must use (E2F) and in larger swarms (E3F). For a swarm whose members are expensive or that must keep working for a long time, robots saved may matter more than items delivered.

Memory is the active ingredient. C1 and C2 differ only in whether a wreck leaves a lasting trace. That difference made C2 robust to detection misses in E7bF, because one sighting is enough to mark a wreck. It also explains why C2 was hurt by a moved hazard in E8F: the old marks remain and point at the wrong place. Faster decay showed no detectable difference.

Alerts that fire did not save robots. C7 with functioning alerts (E10b) lost more robots than C1 in every hazard condition. A candidate explanation is that alerts broadcast the position of a single robot's sighting to all neighbours, causing area avoidance that is rigid in duration rather than graded in strength; C2's marks decay continuously and robots can re-enter once the field is low. The comparison motivates a further variant that allows alert strength to grade like C2's marks, but we do not test it here.

No shared field is needed. In E6bF and E11, keeping marks in each robot's own memory and exchanging them with neighbours showed no detectable difference from a shared field, even with a 0.25 m communication range. This matters for physical robots, which cannot write to a shared virtual field without infrastructure. The result is limited by our arena sizes and our communication model.

Risk weighting. H4F's null result reflects a flat tuning landscape and a gentle risk constant, as under nest restart. A scenario where items lie close to the hazard, or a wider range of k, would be a fairer test.

Limitations. In our main simulator, robots are points without collisions. The Webots replication adds bodies and collisions and keeps the robot-saving effect, but not the small item gain, and it covers only the baseline condition with sensing still provided virtually. All results come from simulation; no physical robots were used. Bit-exact reproducibility holds on Windows; on Linux results agree to within floating-point rounding. There is one hazard in most experiments, which is static except in E8/E8F; the two-hazard results (E12) are a first step toward more complex hazard layouts. Wreck detection is idealised apart from the noise experiments. Robots always know the direction of the nest. The site-fidelity tuning landscape was less flat than under nest restart, but the landscape was not explored beyond the pre-registered grid, and we cannot rule out that another point in parameter space would give stronger results. The site-fidelity replication was not blind (E9 had already revealed its direction). E8F cells 0 and 1 used the nest-tuned decay rather than the fidelity-tuned value, which limits the E8F comparison across decay speeds. E10b was designed after seeing E10 data (disclosed as such).

10 Conclusion

A foraging swarm that cannot sense a hazard can still localise it from the wrecks of its teammates. Under site fidelity, marking delivered more items (median +1 of about 139) and saved robots compared with ignoring wrecks or avoiding only those in view. The robot savings held across swarm sizes, noise levels, moving hazards and two simultaneous hazards. A nest-restart foraging model produced the same ordering, and a physics-based replication with real robot bodies and collisions confirmed the robot savings, although not the small item gain. Risk-limit alerts adapted from the literature that actually fired lost more robots than plain avoidance, while the information medium made no detectable difference. Failures need not only be absorbed; they can also be read.

Code and data availability

The simulator, experiment configurations, analysis and figure scripts, regression tests, raw results and the pre-registration (with its commit history) are publicly available at https://github.com/ayaz-oss/swarm-hazard-marking and archived on Zenodo (DOI 10.5281/zenodo.23591454, release v1.0-preprint). Main experiments ran from commits: fe78c66 (E1), 45b4ce5 (E2), 4cf295a (E3–E5), 5d8ca9e (E6–E7), 37bf380 (E6b, E7b, E8), 50aa7c6 (E9), 7a1ffd6 (tune_c2f), 933e867 (tune_c3f), f8de905 (E1F–E8F), 32e05a6 (tune_c7f), 60c528d (E10–E12), 5f07b60 (E10b) and 2d51e14 (W1; the last 40 W1 runs ran from a1c5de8, which differs only in an analysis script).

Use of AI tools

The simulator and analysis code were written with the help of an AI coding assistant (Claude, Anthropic), under the author's direction. The research question, modelling decisions, pre-registration and interpretation are the author's. Every modelling choice, test and result is documented in the repository.

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Appendix A Deviations from the pre-registration

1. The pilot go/no-go check narrowly missed its threshold (A12 = 0.629 on items delivered against 0.64; robots lost A12 = 0.134). We proceeded by a recorded decision instead of revising the model.
2. Confirmatory tests changed from the unpaired Mann–Whitney U test to the paired Wilcoxon signed-rank test, because runs are paired by seed. This change was made before the main experiments.
3. H3 and H3F were specified as $|A12 - 0.5| < 0.14$ and are operationalised as one-sided bounds (lower CI bound of A12 above 0.36), because the hypotheses concern a cost.
4. The deposit amount was fixed at 1 and the deposit width was tuned instead, because the repellent push scales with deposit amount × repulsion weight.
5. C0 was added to E7 as a reference.
6. A clarification of the E6 confound (how C5 combines deposits) was recorded on 2026-10-19, after the E6/E7 batch had started and before any data were examined.
7. The diagnostics in the original Section 6.8 (nest effect) and E9 were designed after the week-5 results they examine had been seen. E9 was pre-registered before its data existed, but after the effect it tests was identified.
8. **Week-7 replication not blind.** E9 (week 6) already showed, in its fidelity cells, that H1F and H2F would be supported. The pre-registration section for week 7 discloses this explicitly.
9. **E8F decay values.** E8F cells 0 (static) and 1 (moved) used decay = 0.0002 (the nest-tuned value), copied from E8's configuration, rather than the fidelity-tuned decay = 0.002. Only E8F cell 2 used the correct fidelity value. This is a configuration error rather than a deliberate choice, and it limits the E8F across-decay comparison.
10. **E10 tuning flaw.** The pre-registered C7 tuning grid (tune_c7f) used alert_decay = 0.001, which rarely allowed the risk score to reach alert_limit 2 or 3. Mean alerts per run: 0.00 at limit 3, 0.05–0.10 at limit 2, and 3.70–4.55 at limit 1. The pre-registered selection rule chose alert_limit = 3 on noise, making E10's C7 effectively C1. E10 is reported as run.

11. **W1 interruptions.** The W1 batch was paused once by the author and 40 runs hit the 10-minute time limit while the computer was suspended. Only the missing (controller, seed) runs were re-run, with the same seeds and a 30-minute limit; runs that timed out left no data, and no completed run was repeated or discarded.
12. **E10b post-hoc design.** E10b was designed after seeing tune_c7f data (which revealed the tuning flaw) and E10 results. The choice of alert_limit = 1 and alert_duration = 3000 came from the tune_c7f table. The E10b pre-registration section in the hypotheses file discloses this and was committed before any E10b data existed.

Appendix B Additional figures

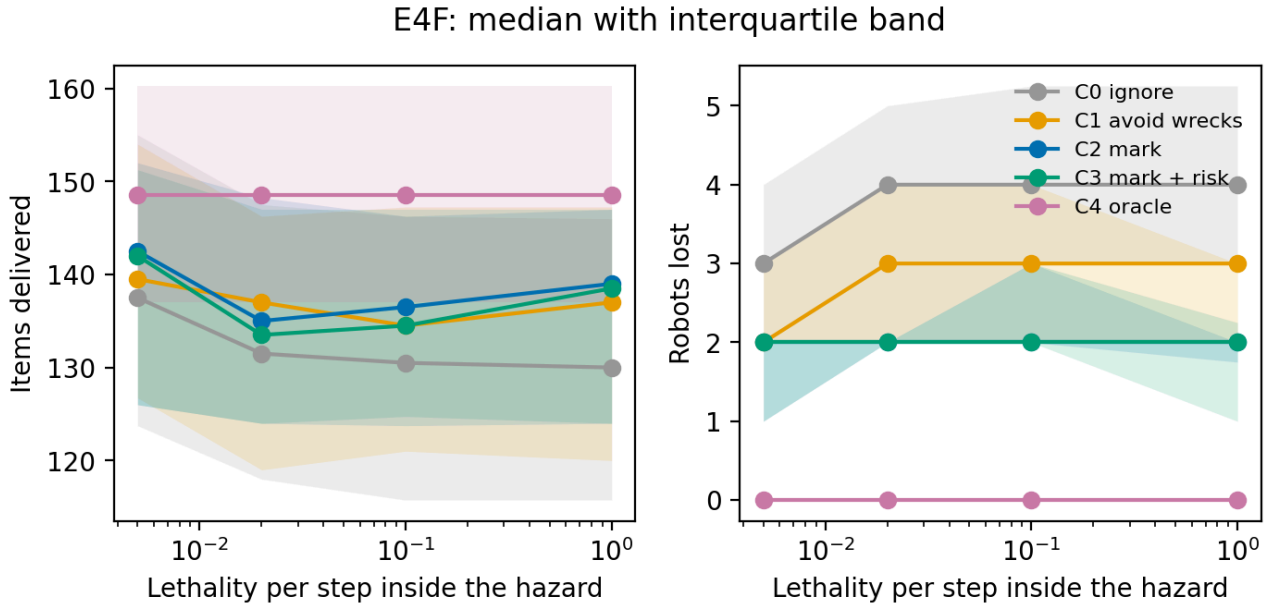


Fig. B1. Lethality under site fidelity (E4F): medians with interquartile bands.

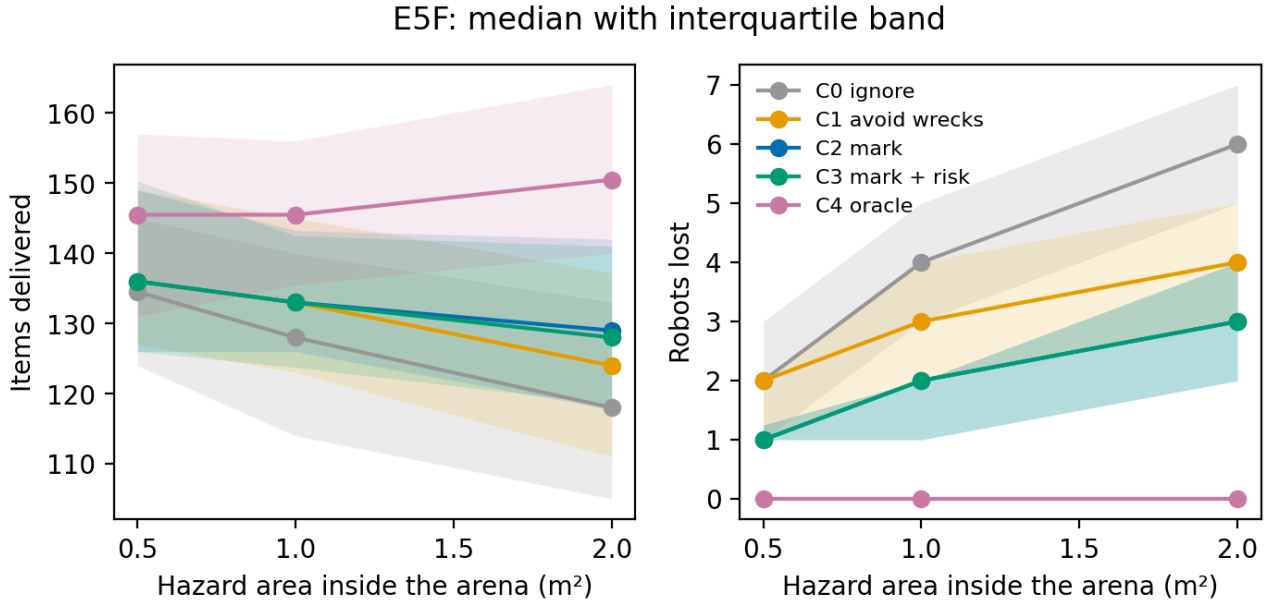


Fig. B2. Hazard size under site fidelity (E5F): medians with interquartile bands against in-arena hazard area.

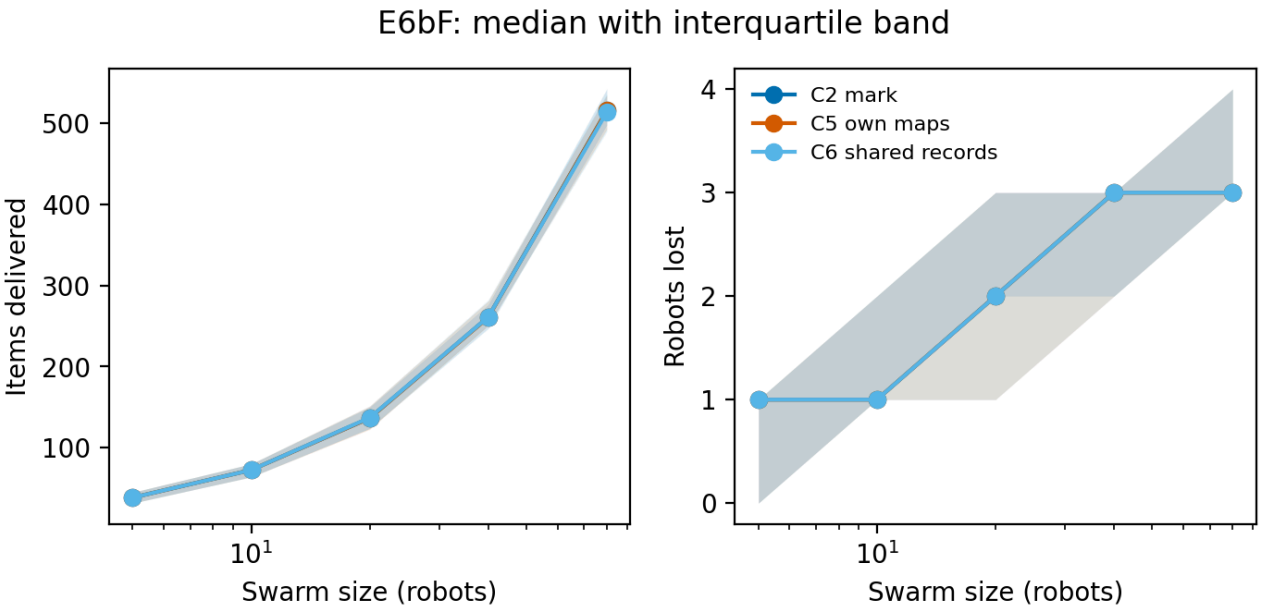


Fig. B3. Information medium under site fidelity (E6bF): C2, C5 and C6 against swarm size.

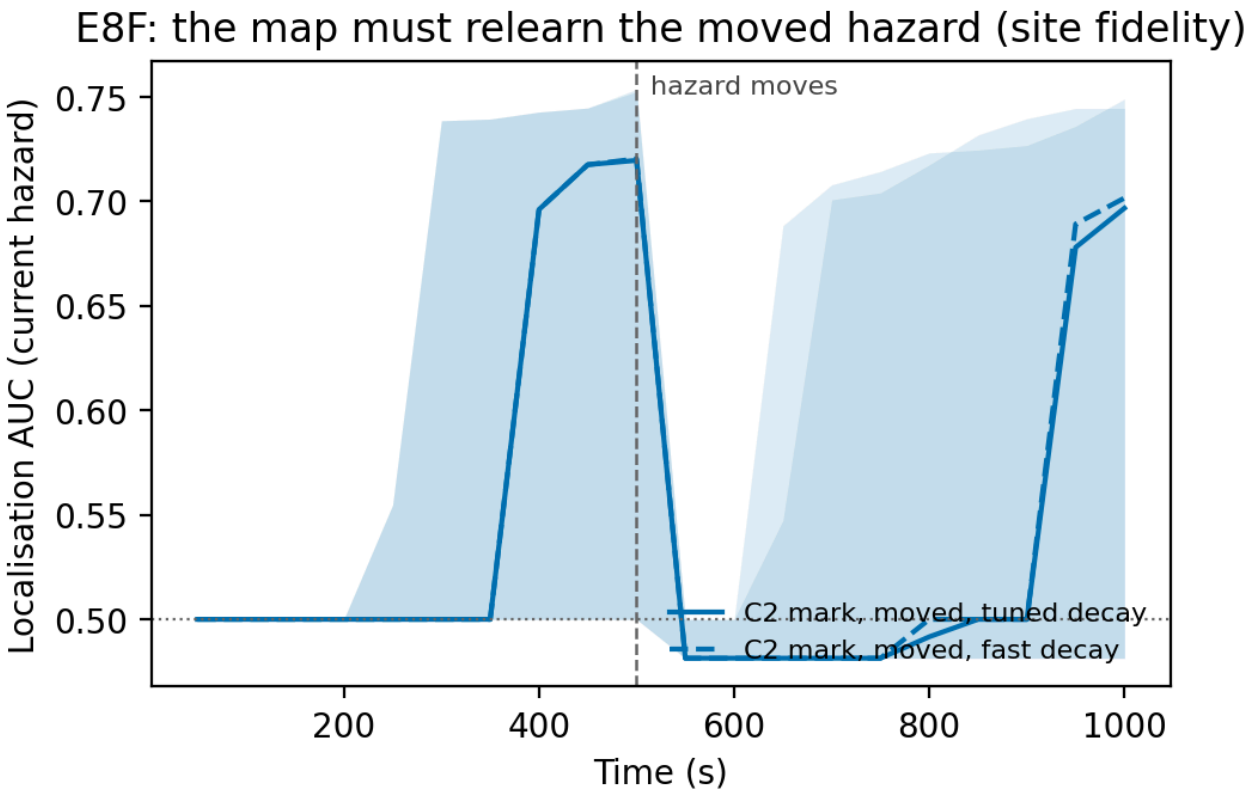


Fig. B4. Localisation AUC over time in E8F (site fidelity), measured against the old hazard before 500 s and the new one after.

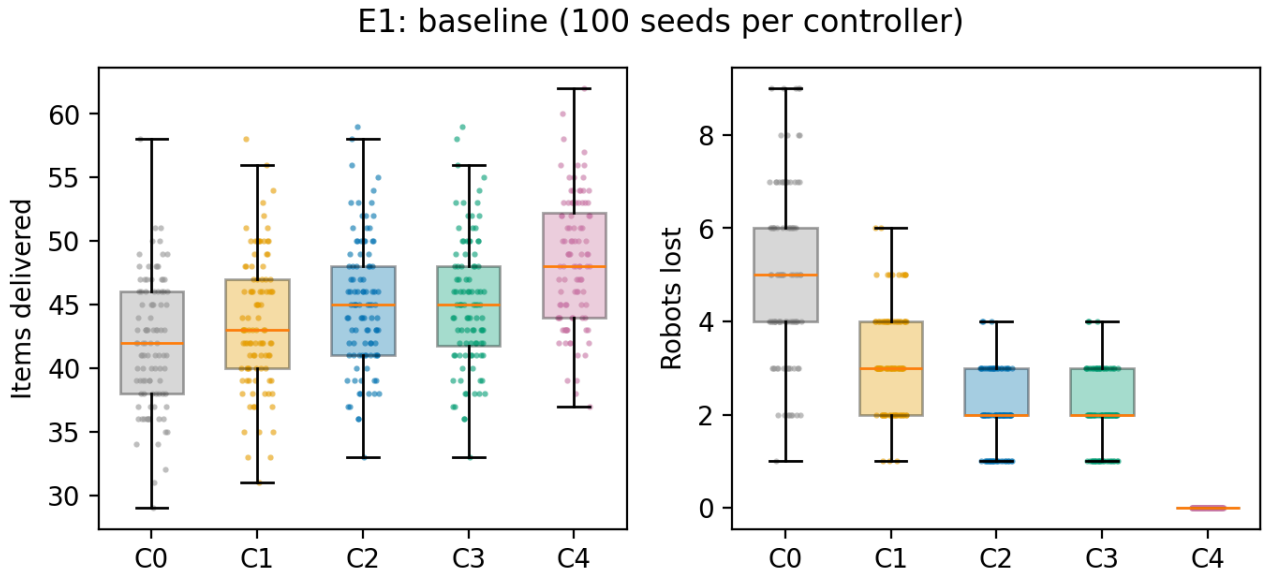


Fig. B5. Baseline under nest restart (E1, 100 seeds), for comparison with Fig. 2.

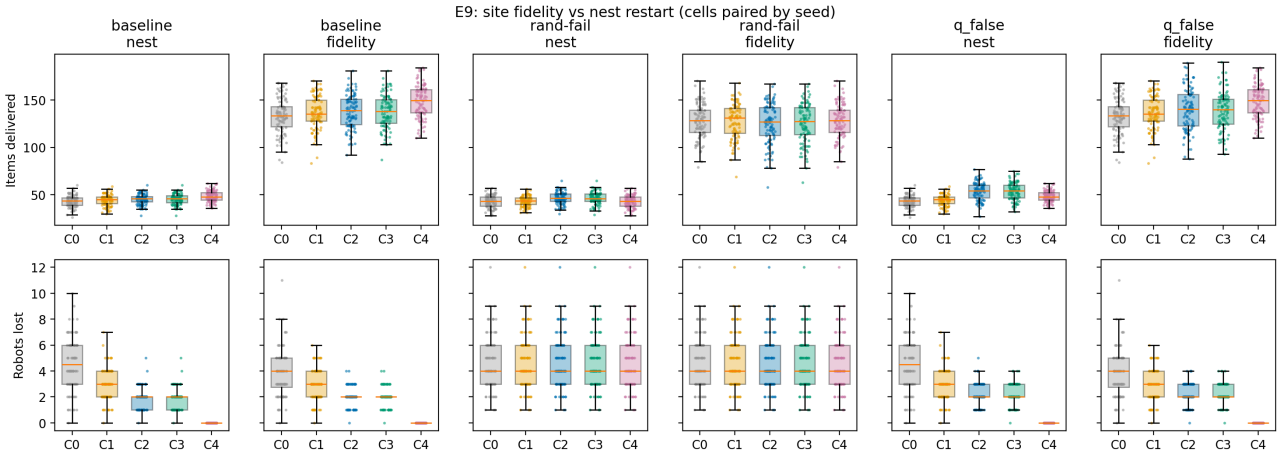


Fig. B6. Site fidelity (E9): items delivered (top) and robots lost (bottom) with search restarting at the nest and with site fidelity, at baseline, in the random-failure control, and with frequent false sightings.